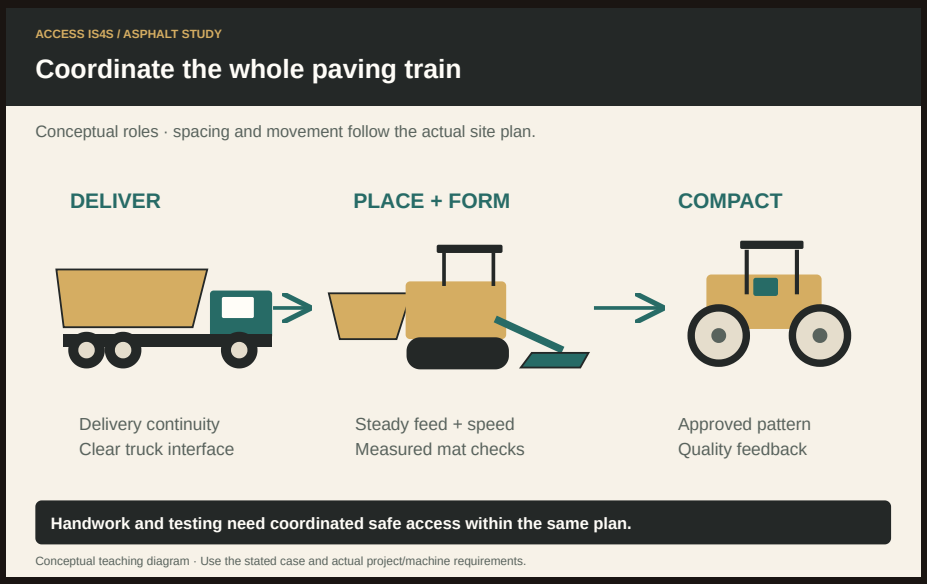
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Paver operation

Volume 6 · I1, I2, I3, I4, I5, I6, I7, I8, I9

Classroom development edition · 30 October 2026



ACCESS IS4S. (2026). Coordinate the whole paving train [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

In this volume

1. How to use the volume and key terms
2. Asphalt placement process
3. Paving machine equipment and attachments
4. Paver work safety procedures
5. Asphalt paver operator roles and responsibilities
6. Paving machine maintenance
7. Markers, grades and stakes
8. Operate asphalt paver
9. Follow shut down procedures
10. Transport paver
11. Keep delivery and mat records consistent

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for paver operation. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS
Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT
Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline I1

Asphalt placement process

Asphalt placement links surface preparation, delivery, paver setup, screed control and rolling. A consistent process helps produce the intended thickness, bond and surface while keeping the whole crew informed.

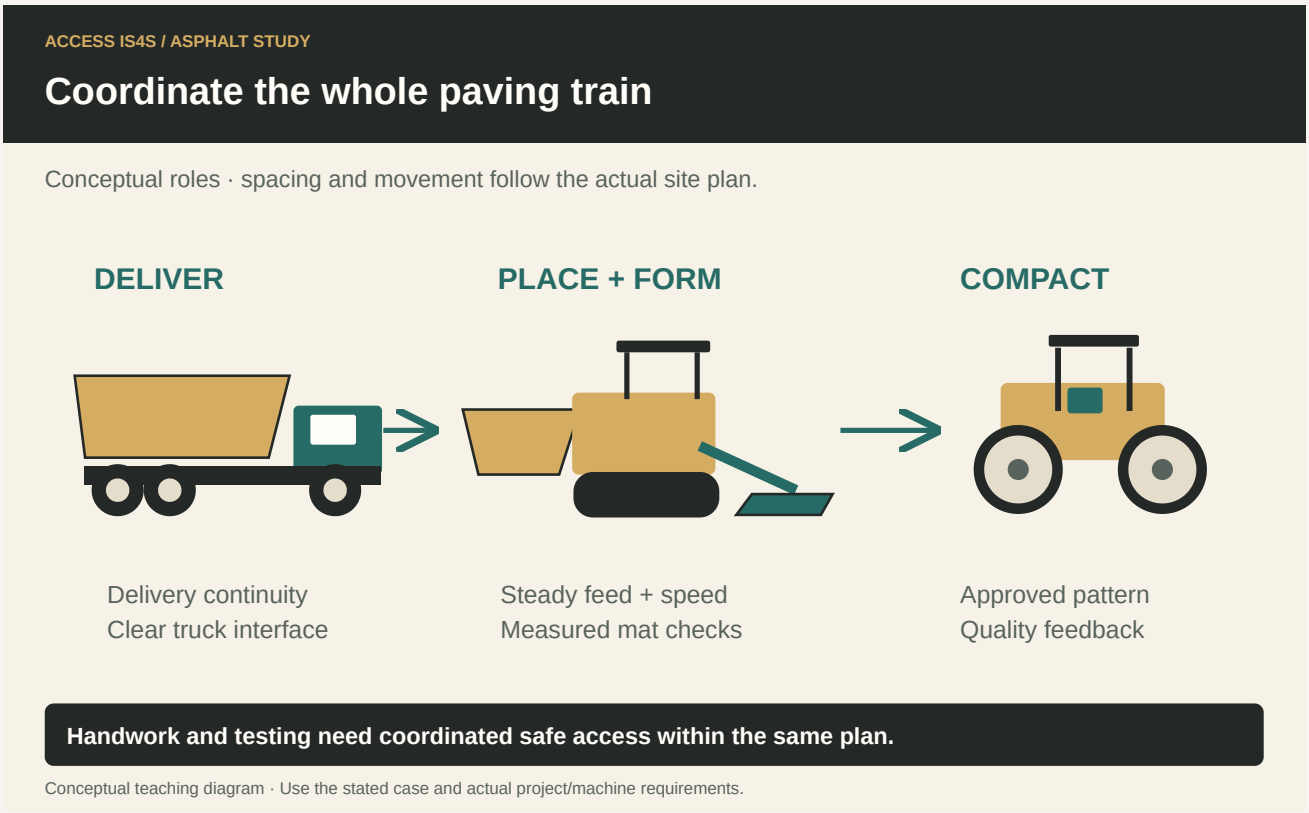
- Place prime and tack coats
- Where to position the Paver
- Ready the Paving Machine
- Lower Screed on to blocks
- Place Haul Truck Loads in Hopper
- Place the asphalt product
- Compact asphalt

LEARNING STEP 1

Place prime and tack coats

Prime and tack have different purposes. A prime treatment may prepare an approved absorbent granular surface; tack promotes bonding between specified pavement surfaces or lifts. Follow the actual product, surface preparation, application rate, curing/breaking and traffic restrictions in the project procedure. Inspect the label and SDS and protect drainage and public routes. Do not substitute one treatment for the other or pave over an unready surface because a truck has arrived.

Coordinate the whole paving train



Note. ACCESS IS4S. (2026). Coordinate the whole paving train [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Where to position the Paver

Position the paver within the planned asphalt train: delivery, placement, handwork and breakdown/intermediate/finish rolling. Confirm line, width, starting joint, access and truck approach. Establish safe separation and communication using the site plan, machine instructions and risk assessment. A diagram of the train explains roles, not universal spacing.

LEARNING STEP 3

Ready the Paving Machine

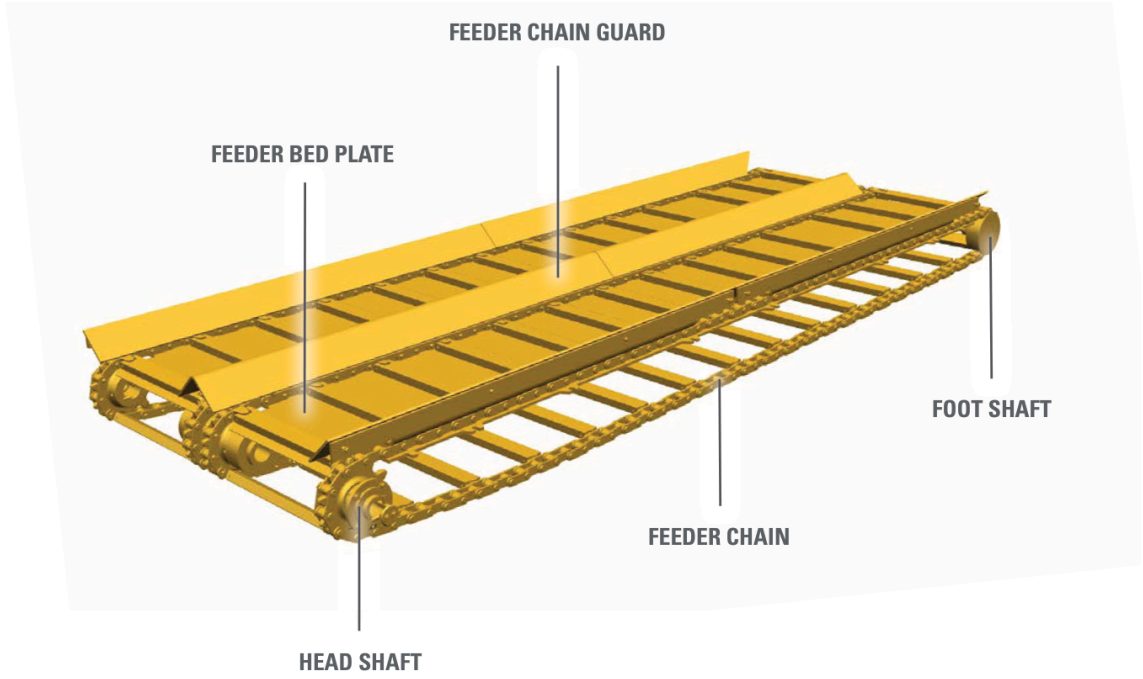
Complete inspection, fluid checks, guards, controls, alarms and the required warm-up. Select the machine's specified operating mode and engine conditions; "full power" is not a universal instruction for every function. Check the fitted feed and grade-control systems and agree who controls each adjustment. Resolve faults before production.

LEARNING STEP 4

Lower Screed on to blocks

Set the screed on the specified starting support/block arrangement and establish initial thickness, crown and extension alignment. Loose thickness must allow for the actual mixture's compaction behaviour and project trial results. A fixed 20% allowance is not universal. Keep hands and feet out of the lowering zone; move or retrieve blocks only through the approved safe sequence.

Inside the paver: the feeder system



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 11). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

LEARNING STEP 5

Place Haul Truck Loads in Hopper

Coordinate the haul truck through the designated signal system. Confirm alignment, clear the pinch zone and control the transfer without striking or disturbing the paver. Hopper-wing operation and load level follow the machine procedure and paving plan. Avoid uncontrolled emptying or repeated material surges. Watch overhead hazards when a truck body rises.

LEARNING STEP 6

Place the asphalt product

Maintain appropriate paving speed and a consistent head of material at the screed. Conveyors carry mix rearward; augers distribute it across the width. The floating screed responds to several interacting factors, so abrupt changes can affect thickness and texture. Plan longitudinal/transverse joints, overlap and tie-ins using the accepted procedure. Record emerging defects and communicate before repeated adjustments.

LEARNING STEP 7

Compact asphalt

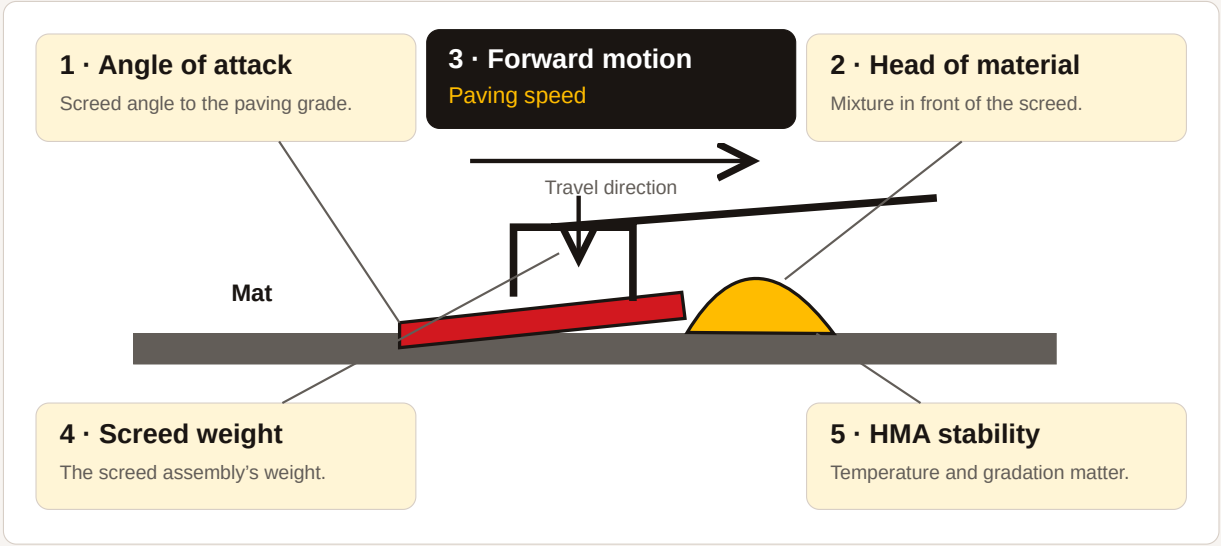
Rolling follows the approved pattern while the mixture is within its effective compaction conditions. Breakdown establishes early compaction; intermediate rolling develops density as required; finish rolling removes marks and completes the surface. Temperature, mixture, support, roller type and weather matter. The paver and roller crews coordinate supply and pace; speed alone does not establish production quality.

A free-floating screed responds to a system

ACCESS IS4S · ASPHALT FOUNDATIONS

A free-floating screed responds to a system

Recognize the five interacting factors before discussing a changed result.



Conceptual redraw · arrows identify relationships, not adjustment instructions or setting values.

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Original instructional adaptation by ACCESS IS4S (2026). Mirrored and simplified the schematic; grouped the five factors in numbered callouts; omitted adjustment procedures and numeric settings. Source locations: [{"figure": "8.16", "printed_page": 383, "pdf_page": 417}, {"section": "8.4.3, five factors", "printed_page": 384, "pdf_page": 418}]. Original contributors: Volvo Construction Equipment Full source context.

Worked example · Convert a trial thickness relationship

A fictional trial produced 50 mm compacted thickness from 62 mm loose thickness. The same approved trial relationship is used to estimate a 45 mm compacted target.

Think it through.

Loose/compacted factor = 62/50 = 1.24. Estimated loose thickness = 45 × 1.24 = 55.8 mm. This is an exercise estimate tied to the stated trial, not a universal setting; confirm with actual placement and compaction checks.

Try it and explain your thinking

Explain and apply the idea

Put surface acceptance, treatment readiness, paver setup, placement and rolling in a useful sequence.

Skills you use: Reading, Problem solving.

A rectangular 50 m × 3 m mat has 0.05 m compacted thickness

A rectangular 50 m × 3 m mat has 0.05 m compacted thickness. Find compacted volume.

Skills you use: Numeracy.

Write a call to the roller lead when delivery is interrupted.

Write a call to the roller lead when delivery is interrupted.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Explain the complete placement sequence and treatment distinction.	0 / 1 / 2	Evidence/help: _____
Verify starting support, line and trial-based thickness allowance.	0 / 1 / 2	Evidence/help: _____
Coordinate delivery, placement and compaction with clear role boundaries.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I1, printed p. 69; source PDF pp. 71–71.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I2

Paving machine equipment and attachments

A paver consists of a tractor that receives and moves material and a screed that forms the mat. Equipment choice depends on the work, access, production and required result—not only the nominal paving width.

Types of paving machines

Major components

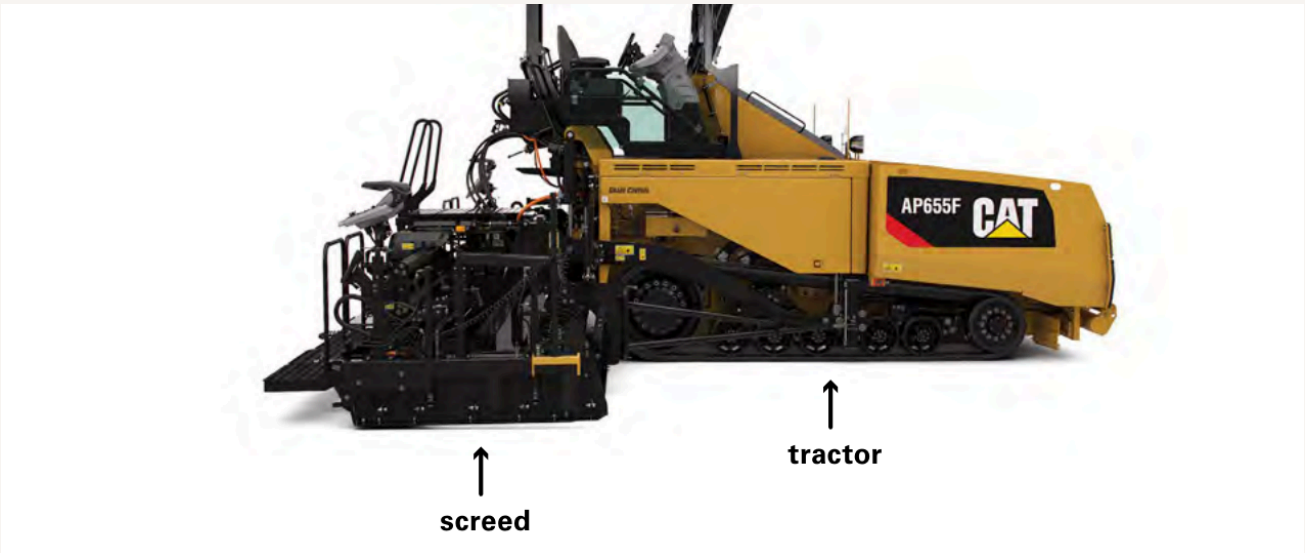
Attachments

LEARNING STEP 1

Types of paving machines

Pavers range from compact pathway or shoulder machines to highway-class equipment. Common nominal categories include 8 ft and 10 ft machines, but actual minimum/maximum widths, extensions and transport dimensions depend on the model. Compare work width, production, turning space, support, transport and fitted options. Light, medium and highway duty describe broad applications rather than interchangeable operating limits. Match the OMM to model, serial range and configuration.

Two main sections of the paver



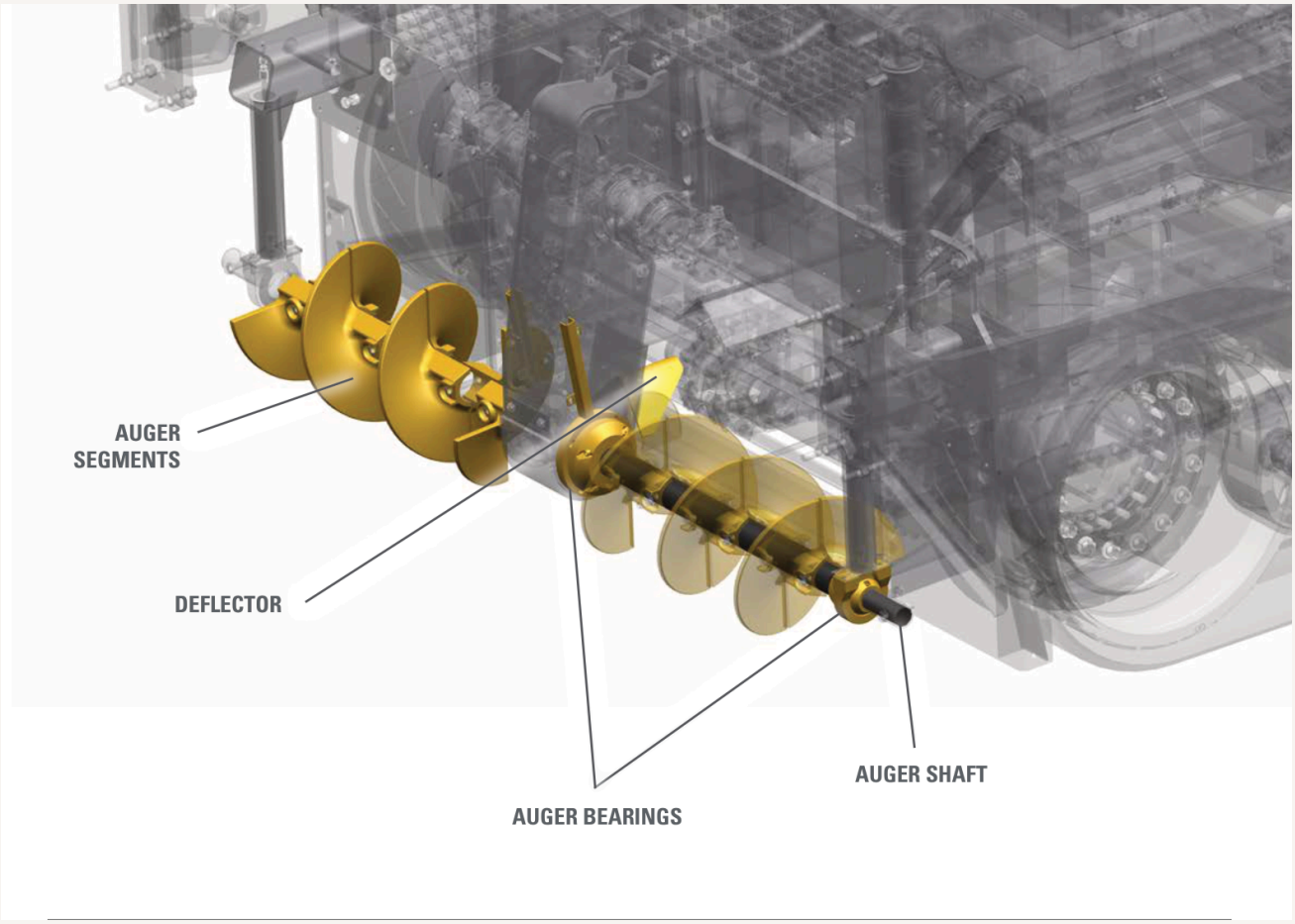
Note. Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888, source PDF p. 63). Faithful crop; original labels retained. Conceptual paver example in the 2016 application guide; this picture does not identify a specific machine serial number. Full source context.

LEARNING STEP 2

Major components

Identify the tractor's hopper and wings, push rollers/truck interface, slat conveyors or other fitted feed system, augers, engine, hydraulics, undercarriage, operator station and grade/slope controls. Wheeled, rubber-tracked and steel-tracked arrangements differ. Identify the screed's tow arms/tow points, main plate, extensions, end gates, heating and vibration/tamper equipment where fitted. Conveyors move material toward the screed; augers spread it; the screed forms the layer. Do not confuse a feed sensor with a grade sensor.

Auger parts: identify the assembly



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 15). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

LEARNING STEP 3

Attachments

A pickup machine can collect a windrow placed by an appropriate delivery system and feed the paver. A material-transfer vehicle or hopper insert may buffer supply or remix material, depending on design. These can reduce certain handling problems but do not guarantee freedom from segregation or a universal productivity increase. Belly-dump and end-dump arrangements require different work plans. Check compatibility, added weight, clearance and operating instructions for every attachment.

Feed sensing and hopper information

SONIC MATERIAL FEED SENSORS

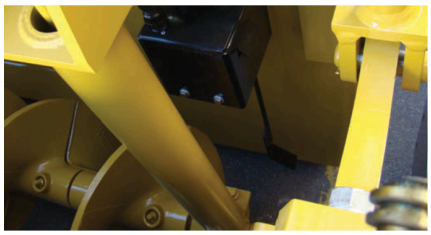
Four sonic transducers mounted above the conveyor discharge and at the end of the augers activate automatically to maintain a preset amount of material ahead of the screed.

Two sonic transducers mounted at the end of the augers automatically maintain a preset amount of material ahead of the screed.



PADDLE MATERIAL FEED SENSORS

Two paddle sensors located in the auger chamber activate the conveyors to automatically maintain the desired amount of material ahead of the screed.



HOPPER MATERIAL LEVEL AND TEMPERATURE SENSOR

Mounts above the hopper to monitor the amount of material and the temperature of the mix using a single infrared temperature sensor. Material level and temperature information is available on the machine display at the operator station.

Not compatible with certain serial numbers. Before ordering, provide machine serial number to your Cat dealer.



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 43). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · Two sensor jobs

A diagram labels one sensor near the auger and another following a reference beside the machine.

Think it through.

The feed sensor can control the material supply in its assigned zone; a grade sensor follows a reference for elevation control. Confirm the actual model and function from the manual. Similar-looking sensors do not necessarily control the same system.

Try it and explain your thinking

Label ten parts on the supplied paver figure and state each function.

Label ten parts on the supplied paver figure and state each function.

Skills you use: Reading, Communication.

Convert nominal 8 ft and 10 ft widths to metres

Convert nominal 8 ft and 10 ft widths to metres. Do these establish the actual screed settings?

Skills you use: Numeracy, Problem solving.

Ask an AI to create a component table from the supplied figure and caption

Ask an AI to create a component table from the supplied figure and caption. How will you audit it?

Skills you use: Digital, Writing, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Identify tractor and screed assemblies accurately.	0 / 1 / 2	Evidence/help: _____
Distinguish feed, grade and slope-control functions.	0 / 1 / 2	Evidence/help: _____
Check attachment compatibility and document scope.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I2, printed p. 70; source PDF pp. 72–72.
- Caterpillar. (Various editions). Original machine component figures, with publication/page/model records. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I3

Paver work safety procedures

Safe paver work depends on a shared plan and controlled movement around hot material and moving machinery. The ground crew, truck driver and screed crew need the same understanding of the work boundary.

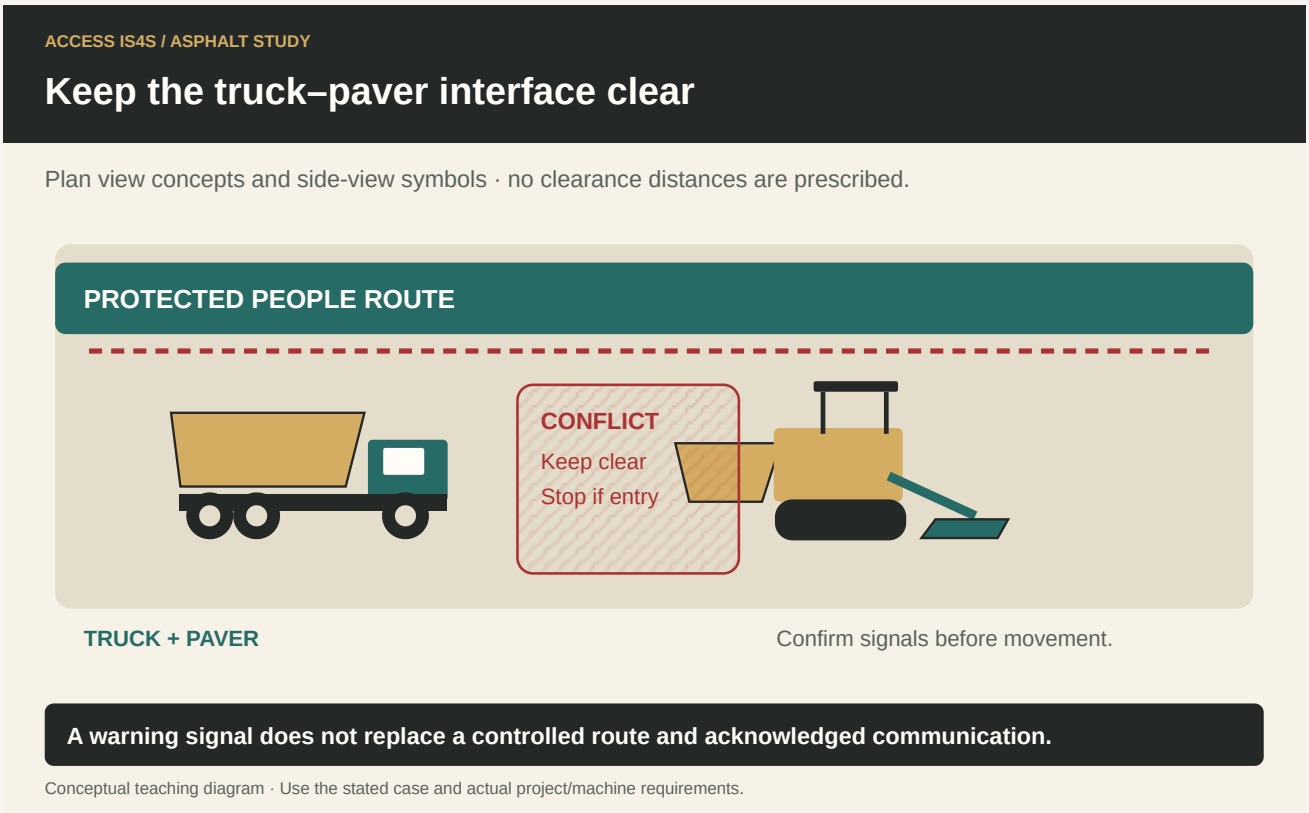
- Interpret job plan
- Awareness of machine-specific safety
- Traffic control requirements
- Use proper PPE
- Lockout procedures
- Secure machine for safety

LEARNING STEP 1

Interpret job plan

Brief the path, paving pattern, thickness, crown, joints, obstacles, utilities and public access. Identify the approved drawing and changes. Establish who directs truck movement, who controls the screed and how an emergency stop is communicated. Protect entrances and services according to the actual access plan.

Keep the truck–paver interface clear



Note. ACCESS IS4S. (2026). Keep the truck–paver interface clear [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Awareness of machine-specific safety

Recognize hopper-wing and truck-interface pinch points, conveyor/auger entanglement, hot mix and heating systems, fumes, noise, hydraulic pressure, blind areas and moving rollers. Never enter a hopper or reach into a feed system because motion has paused. Keep clear of the machine's swept path and raised components. Use specified blocking/support before work requiring access.

LEARNING STEP 3

Traffic control requirements

Use the traffic-control plan for signs, devices, trained personnel and communication. Public traffic control and guiding a delivery truck are distinct roles. Reassess as the paver advances and access points change. Stop affected movements when the route or signals become unclear; do not assume high-visibility clothing makes a worker visible to every operator.

LEARNING STEP 4

Use proper PPE

Use the required footwear, head, eye, hearing, visibility and heat/chemical protection selected for the task. Check condition and fit. Respiratory protection needs the applicable program and controls. The employer ensures required PPE is used and maintained; worker/employer supply responsibilities follow current Part 8 and any agreements. PPE does not remove a crush or entanglement hazard.

LEARNING STEP 5

Lockout procedures

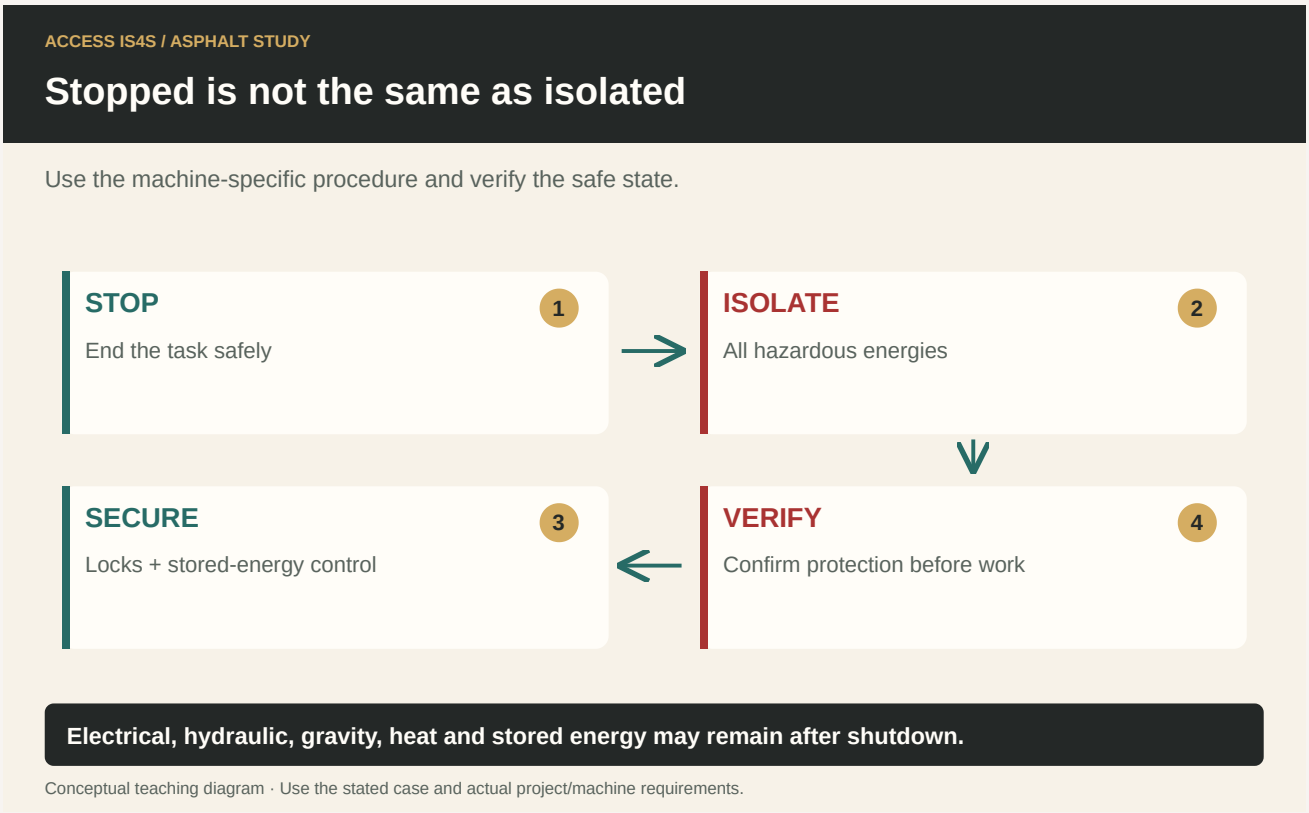
Apply the machine-specific lockout procedure for service, clearing or hazardous access. Address engine/electrical, hydraulic, gravity, heat and stored energy. Secure and verify the safe state before work. A night switch, stopped engine or emergency-stop button alone is not proof of complete isolation. Reinstatement follows the authorized procedure with people clear and guards restored.

LEARNING STEP 6

Secure machine for safety

Park at the designated stable location, position and support the screed/attachments as instructed, apply parking controls and required chocks, and secure access. Record defects and communicate out-of-service status. Protect people who may approach after the crew leaves.

Stopped is not the same as isolated



Note. ACCESS IS4S. (2026). Stopped is not the same as isolated [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A stopped auger

Material stops moving, and a worker reaches toward the auger to remove a lump.

Think it through.

Stop the approach and keep clear. A pause in automatic feed does not isolate the auger; it may restart. Use the approved stopping, lockout, stored-energy and clearing procedure with authorized personnel.

Try it and explain your thinking

Name the safe response when the truck guide’s signal is lost.

Name the safe response when the truck guide’s signal is lost.

Skills you use: Communication, Collaboration.

A revised layout moves the paver edge 0.4 m toward a public route

A revised layout moves the paver edge 0.4 m toward a public route. What must be checked before proceeding?

Skills you use: Numeracy, Problem solving, Adaptability.

Prepare a three-field defect note for a damaged hopper guard.

Prepare a three-field defect note for a damaged hopper guard.

Skills you use: Writing, Digital.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use selected PPE and identify machine-specific hazards.	0 / 1 / 2	Evidence/help: _____
Demonstrate full approved lockout and verification.	0 / 1 / 2	Evidence/help: _____
Secure the machine with required supports, parking controls and access protection.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I3, printed p. 71; source PDF pp. 73–74.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 18: Traffic Control. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I4

Asphalt paver operator roles and responsibilities

The paver operator maintains alignment and a steady process while coordinating the truck and screed crews. Good operation includes noticing a developing problem early and communicating it clearly.

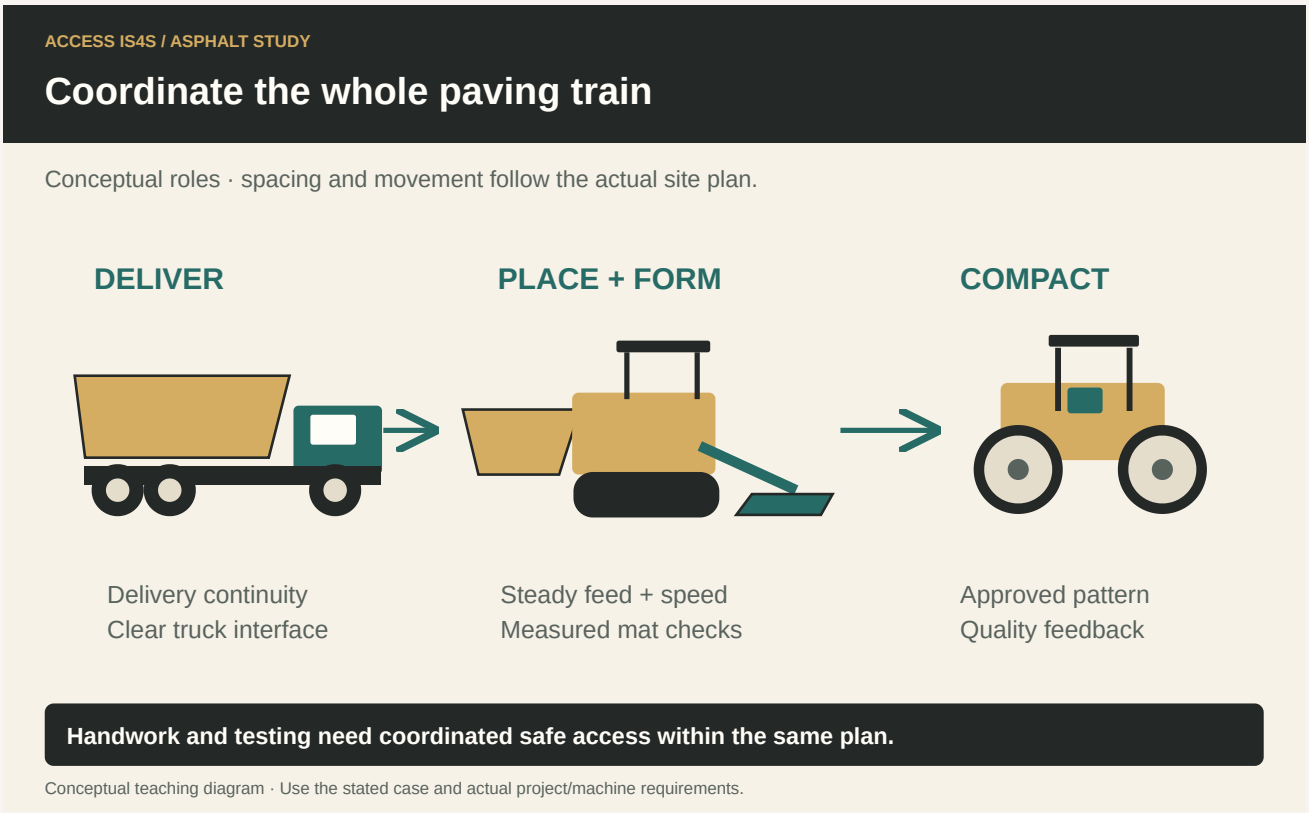
- Position Paver
- Ready the Paver
- Follow the line
- The transition process
- Maintain consistent “head”
- Positioning of haul trucks
- Signals/communication
- Obstacles and overhead hazards

LEARNING STEP 1

Position Paver

Position the paver to the approved starting line and joint with sufficient working room. Check the first pass carefully because it may guide later passes. Avoid unnecessary manoeuvres and abrupt corrections. Confirm the ground support and the whole machine’s clearance, including extensions and attachments.

Coordinate the whole paving train



Note. ACCESS IS4S. (2026). Coordinate the whole paving train [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Ready the Paver

Complete prechecks, appropriate warm-up and screed preparation. Heating prevents material sticking to cold plates; overheating can damage the screed. Follow the fitted heating system's procedure and verify ready status. Set starting supports and thickness through the agreed screed setup; never use an assumed generic block size.

LEARNING STEP 3

Follow the line

Follow the authorized reference, which may be stringline, a suitable edge, laser or other grade/alignment system. Look ahead and make smooth steering changes. Check that the reference remains valid and protected. Automatic controls can follow an incorrect or disturbed reference faithfully, so independent checks remain necessary.

LEARNING STEP 4

The transition process

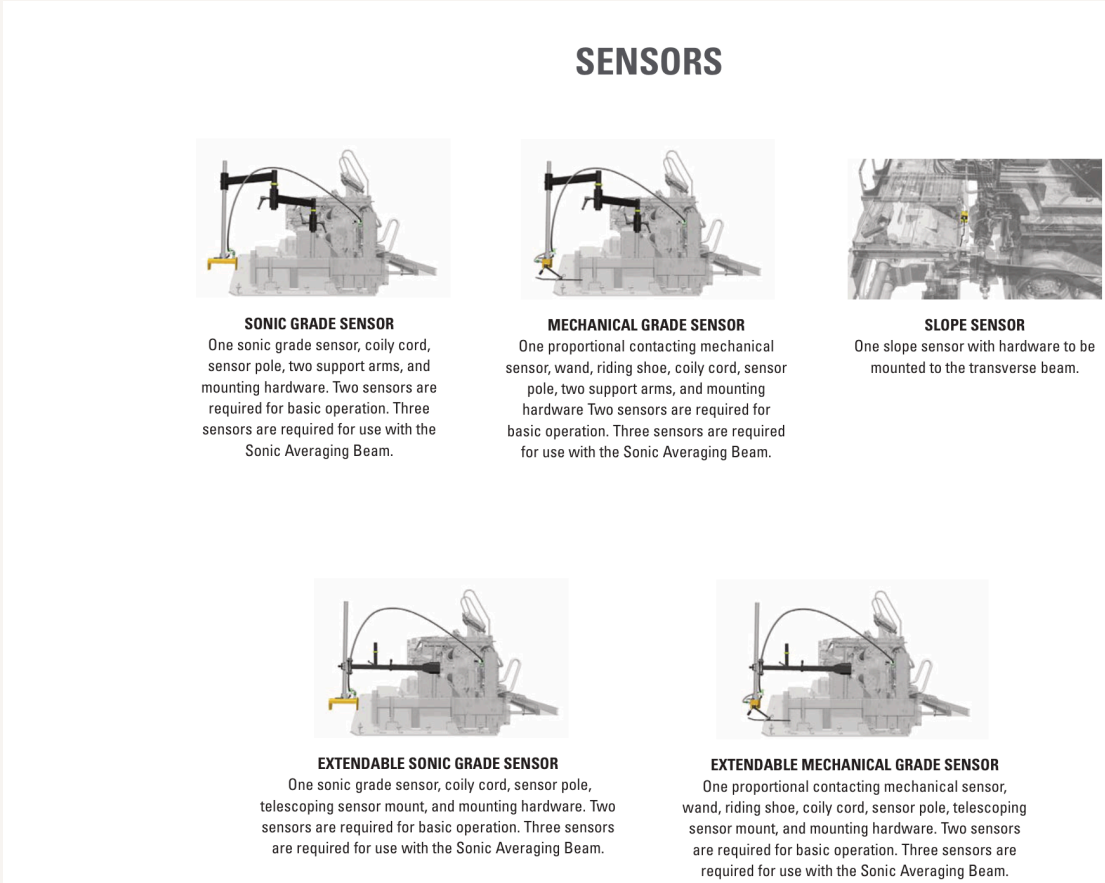
Coordinate truck changes before the hopper starves or receives a sudden surge. The delivery rate, available buffer, paver speed and planned stopping method must work together. Communicate delays early. Follow the machine's hopper-wing and feed procedure rather than repeatedly cycling wings as a universal habit.

LEARNING STEP 5

Maintain consistent "head"

A consistent head of material in front of the screed supports stable operation. It is the controlled quantity/distribution at the screed, not simply a completely full hopper. Monitor feed sensors, conveyors and augers as fitted. Changes in head, speed and mixture conditions can alter the mat. Coordinate corrective action with the screed operator.

Identify grade and slope sensors



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 34). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

LEARNING STEP 6

Positioning of haul trucks

Guide haul trucks into the approved alignment using the designated signalling arrangement. Keep people clear of the truck/paver interface and verify clearance before unloading. Avoid impacts that disturb the paver. Raised bodies change overhead clearance and stability considerations; the driver follows the vehicle procedure and site plan.

LEARNING STEP 7

Signals/communication

Use agreed, deliberate signals that the intended driver can see through the appropriate mirror or communication system. Obtain acknowledgement. A horn’s meaning must be agreed, not guessed. One designated person directs the movement; anyone can call an emergency stop. Lost or conflicting signals mean stop.

LEARNING STEP 8

Obstacles and overhead hazards

Scan the route and overhead envelope for utilities, structures, branches, attachments, other machines and people. Account for the truck body, hopper wings and screed extensions as they move. Electrical approach limits depend on the actual system and required controls; do not substitute a guessed distance. Escalate an unresolved obstacle before entering its zone.

Feed sensing and hopper information

SONIC MATERIAL FEED SENSORS

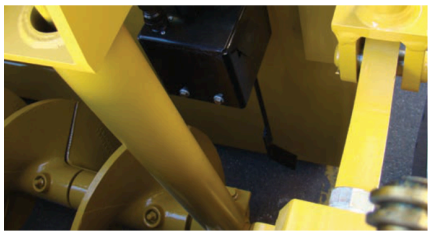
Four sonic transducers mounted above the conveyor discharge and at the end of the augers activate automatically to maintain a preset amount of material ahead of the screed.

Two sonic transducers mounted at the end of the augers automatically maintain a preset amount of material ahead of the screed.



PADDLE MATERIAL FEED SENSORS

Two paddle sensors located in the auger chamber activate the conveyors to automatically maintain the desired amount of material ahead of the screed.



HOPPER MATERIAL LEVEL AND TEMPERATURE SENSOR

Mounts above the hopper to monitor the amount of material and the temperature of the mix using a single infrared temperature sensor. Material level and temperature information is available on the machine display at the operator station.

Not compatible with certain serial numbers. Before ordering, provide machine serial number to your Cat dealer.



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 43). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · Hopper level is not the whole story

The hopper contains mix, but the right side of the screed has a visibly reduced material head and texture changes.

Think it through.

Report the side and location, coordinate with the screed operator and inspect the feed-system status from safe positions. A full hopper does not prove uniform delivery across the screed. Use authorized adjustments and record the response; do not reach into the feed system.

Try it and explain your thinking

Explain the difference between hopper contents and head of material.

Explain the difference between hopper contents and head of material.

Skills you use: Reading, Communication.

A 20 t load lasts 8 min at a steady fictional rate

A 20 t load lasts 8 min at a steady fictional rate. Find t/min; why is it not a universal paver speed?

Skills you use: Numeracy, Problem solving.

Draft a truck-change handover that includes an access constraint.

Draft a truck-change handover that includes an access constraint.

Skills you use: Writing, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Align and follow a verified reference.	0 / 1 / 2	Evidence/help: _____
Coordinate truck transitions and consistent material head.	0 / 1 / 2	Evidence/help: _____
Recognize and respond to obstacles and communication loss.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I4, printed p. 73; source PDF pp. 75–75.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 3: Rights and Responsibilities. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I5

Paving machine maintenance

A clean, correctly serviced paver feeds material consistently and gives the crew safe access. Daily maintenance follows the selected machine's schedule and safe conditions, with defects recorded for the next shift.

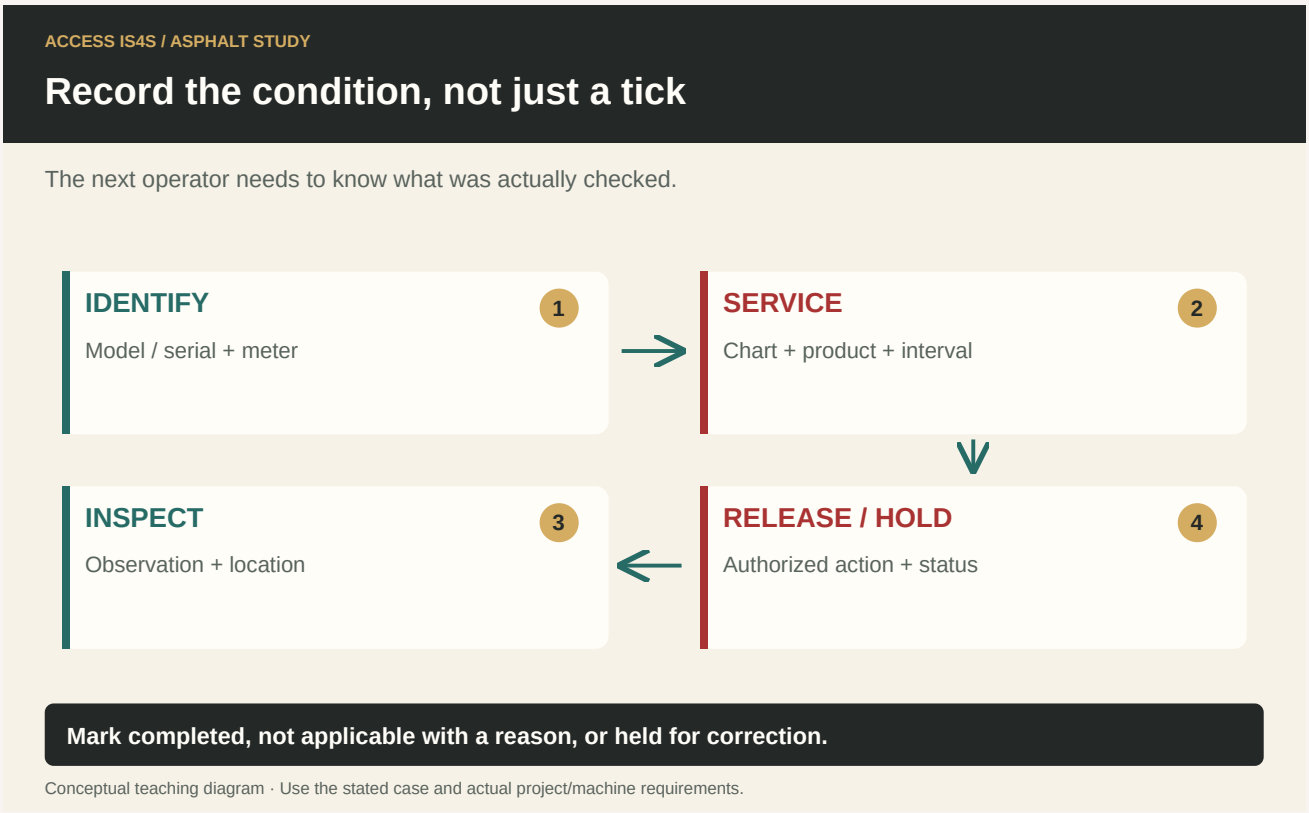
- Service lubrication system (greasing)
- Service air intake system as required (air filters/pre cleaners)
- Drain fuel tank sump, water separator fuel filters if equipped
- Add fluids as required
- Inspect and clean components as required
- Housekeeping tasks

LEARNING STEP 1

Service lubrication system (greasing)

Locate lubrication points on the actual chart, including fitted centralized systems. Confirm grease type, interval, quantity and safe position. Clean fittings and use a properly loaded grease gun. The LeeBoy 8680 example includes optional systems that combine several points; do not assume every machine has them. Record service and report damaged or blocked fittings instead of forcing lubricant through an unknown obstruction.

Record the condition, not just a tick



Note. ACCESS IS4S. (2026). Record the condition, not just a tick [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Service air intake system as required (air filters/pre cleaners)

Read the intake restriction indicator, inspect the pre-cleaner and check housing/seal condition. Follow the manufacturer’s cleaning/replacement policy and keep dirt out of the clean side. Some elements must be replaced rather than cleaned. Do not use compressed air unless the actual procedure permits it and specifies the method and limits.

LEARNING STEP 3

Drain fuel tank sump, water separator fuel filters if equipped

Service the fuel sump or separator only where fitted and under its prescribed conditions. Collect drainage, note water/debris and dispose of it properly. Keep ignition sources controlled. Routine separator service does not authorize opening high-pressure fuel components. Repeated contamination requires investigation.

LEARNING STEP 4

Add fluids as required

Match fluid to the labelled system and specification. Check ground level, temperature and stopped/running condition before interpreting the dipstick or sight gauge. Use the correct side of any dual-scale dipstick. Clean fill points, avoid cross-contamination and overfilling, and record unusual consumption. Coolant, hydraulic fluid, engine oil and release agent are not interchangeable.

LEARNING STEP 5

Inspect and clean components as required

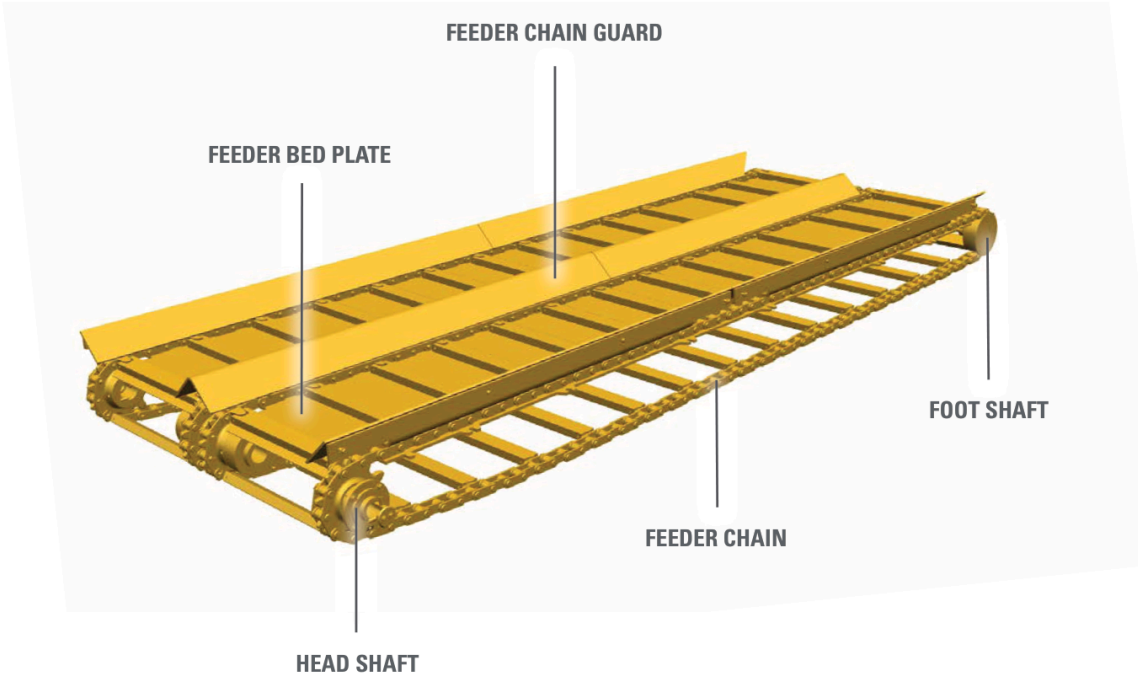
Inspect radiator/oil cooler, accessible plugs, dipsticks, battery condition and hoses using safe access and isolation. Protect cooling fins and electrical parts during cleaning. Never open a hot pressurized system or touch a suspected hydraulic leak. Restore guards and refer repairs to authorized personnel.

LEARNING STEP 6

Housekeeping tasks

Clean hopper, conveyors, augers, screed decks, extensions, steps and commonly missed ledges using the approved procedure. Coordinate with the screed crew, control heat and ignition sources, and prevent contact with moving systems. Use a compatible release/cleaning product with its SDS and containment requirements. “Biodegradable” does not mean harmless or permitted to drain onto the ground. Collect waste, account for tools and document the final condition.

Inside the paver: the feeder system



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 11). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · A repeated low level

Hydraulic oil is low again after yesterday’s top-up. A fresh wet area is visible beneath the machine.

Think it through.

Record the repeated loss and safe visual observation, secure/hold the machine as required and notify the responsible person. Do not treat repeated top-up as a repair or touch the suspected leak. Confirm authorized correction before release.

Try it and explain your thinking

A fictional service interval is 10 operating hours

A fictional service interval is 10 operating hours. Last service 1,420 h; current 1,429.5 h. How much remains?

Skills you use: Numeracy, Reading.

Explain why an unlabelled spray bottle cannot be accepted as a release agent.

Explain why an unlabelled spray bottle cannot be accepted as a release agent.

Skills you use: Problem solving, Communication.

Create a digital maintenance record without claiming checks you did not perform.

Create a digital maintenance record without claiming checks you did not perform.

Skills you use: Digital, Writing, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Service lubrication/intake/separator according to the actual schedule.	0 / 1 / 2	Evidence/help: _____
Check correct fluids and accessible cooling/battery components safely.	0 / 1 / 2	Evidence/help: _____
Clean feed and access areas with containment and restore guards.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I5, printed p. 74; source PDF pp. 76–77.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- Lincoln Industrial. (December 2013; Form 404636). Lever Gun Models 1142, 1147 and 1151: Installation and Maintenance Guide. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I6

Markers, grades and stakes

A paver operator uses survey information to maintain the approved line and geometry. The marks must be interpreted before they become machine instructions.

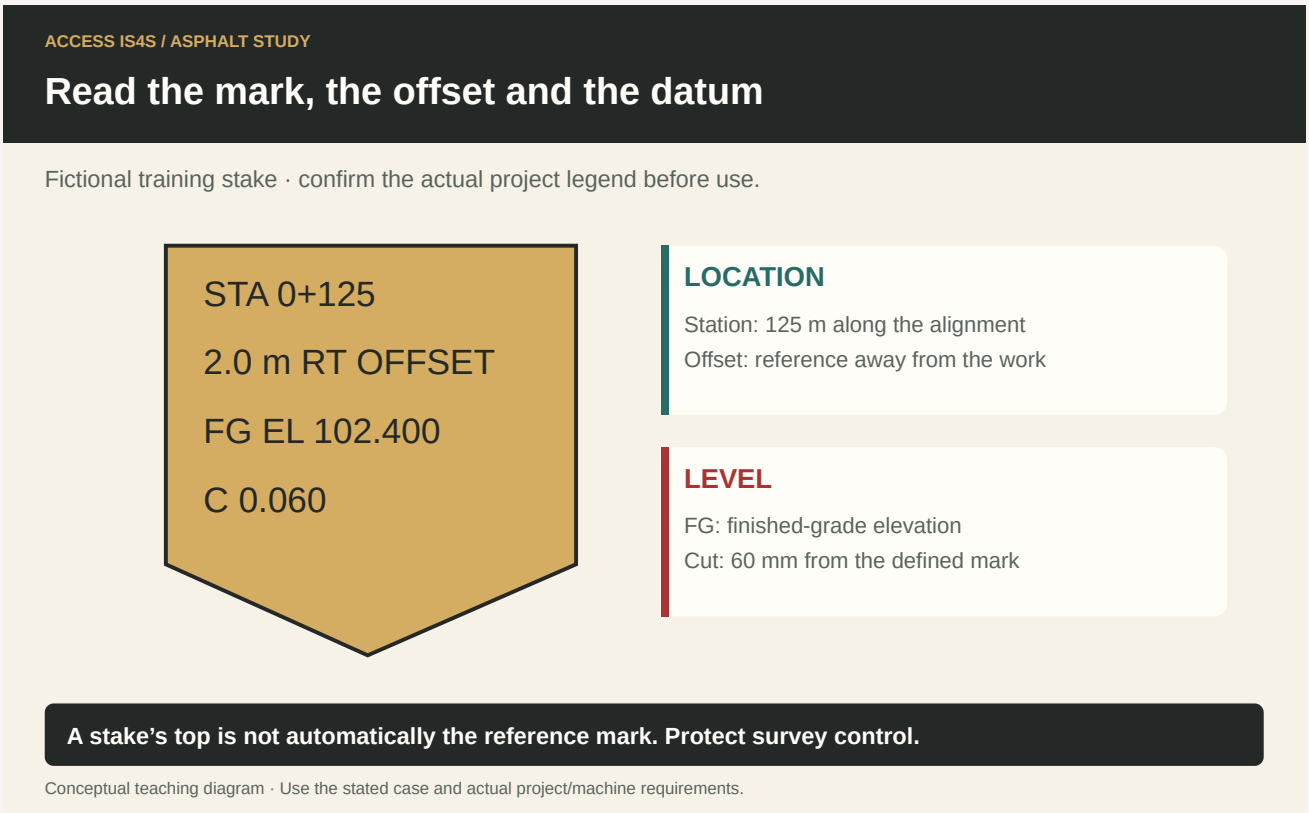
- Abbreviations used on stakes
- Interpret stakes
- Fundamentals of paver operation

LEARNING STEP 1

Abbreviations used on stakes

Read common abbreviations with the project legend: BM benchmark; C cut; CB catch basin; SMH sanitary manhole; EL elevation; FG finished grade; PL property line; RW right of way; SAN/S sanitary; Mon monument; G gas; PVC polyvinyl chloride. A label identifies a feature or value, not necessarily its complete location or safe clearance. Keep units and datum with every elevation.

Read the mark, the offset and the datum



Note. ACCESS IS4S. (2026). Read the mark, the offset and the datum [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Interpret stakes

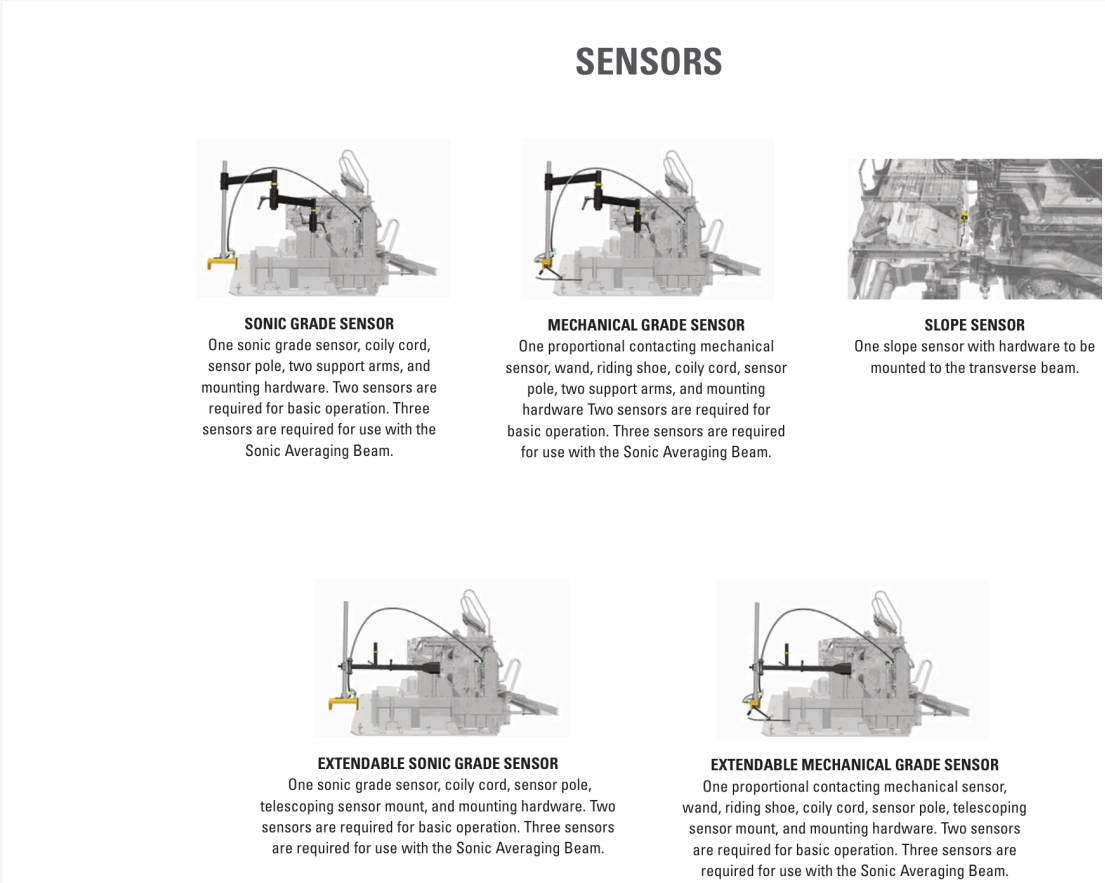
Identify station, side, offset, reference mark, cut/fill, slope and target work. Preserve property pins, benchmarks and protected monuments. Distinguish an assumed local datum from geodetic control. Cross-heads and offset stakes transfer information away from the work area; they are not automatically the paving edge. If a mark is moved or conflicts with the current drawing, stop using it and request verification.

LEARNING STEP 3

Fundamentals of paver operation

Follow a verified line with a stable sighting reference and smooth steering. Look sufficiently ahead while maintaining awareness of trucks, crew and obstacles. Guide trucks into the planned straight approach rather than forcing the paver out of alignment to receive them. Experience improves precision through feedback, not by ignoring measured checks. Coordinate alignment with the screed operator's width and grade controls.

Identify grade and slope sensors



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 34). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · Offset versus edge

A stake marked 1.5 m RT OFFSET describes the approved right paving edge at station 0+060. A learner treats the stake itself as the edge.

Think it through.

Use the project legend and stated offset direction to locate the actual edge; the stake is a reference away from it. Verify the relation with the grade lead before setting the paver line. Do not shift the whole mat to the stake.

Try it and explain your thinking

At one verified point, base EL is 101.250 m and finished EL is 101.300 m

At one verified point, base EL is 101.250 m and finished EL is 101.300 m. Find the vertical difference.

Skills you use: Numeracy, Problem solving.

Write a read-back containing station, side, offset and unresolved datum.

Write a read-back containing station, side, offset and unresolved datum.

Skills you use: Writing, Communication.

Communicate and verify the work

Use an AI to explain the abbreviation list, then identify what it cannot establish from the list alone.

Skills you use: Digital, Reading, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Interpret and protect the actual control marks.	0 / 1 / 2	Evidence/help: _____
Transfer line/elevation information correctly.	0 / 1 / 2	Evidence/help: _____
Maintain smooth alignment while coordinating truck and screed roles.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I6, printed p. 76; source PDF pp. 78–79.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Leica Geosystems. (n.d.). Leica NA700 Series Automatic Levels: Technical Data. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I7

Operate asphalt paver

Paver operation is learned through observed, supervised control of the actual machine. In-school practical work in the outline is on flat surfaces; workplace performance adds the approved grades, slopes and conditions under competent supervision.

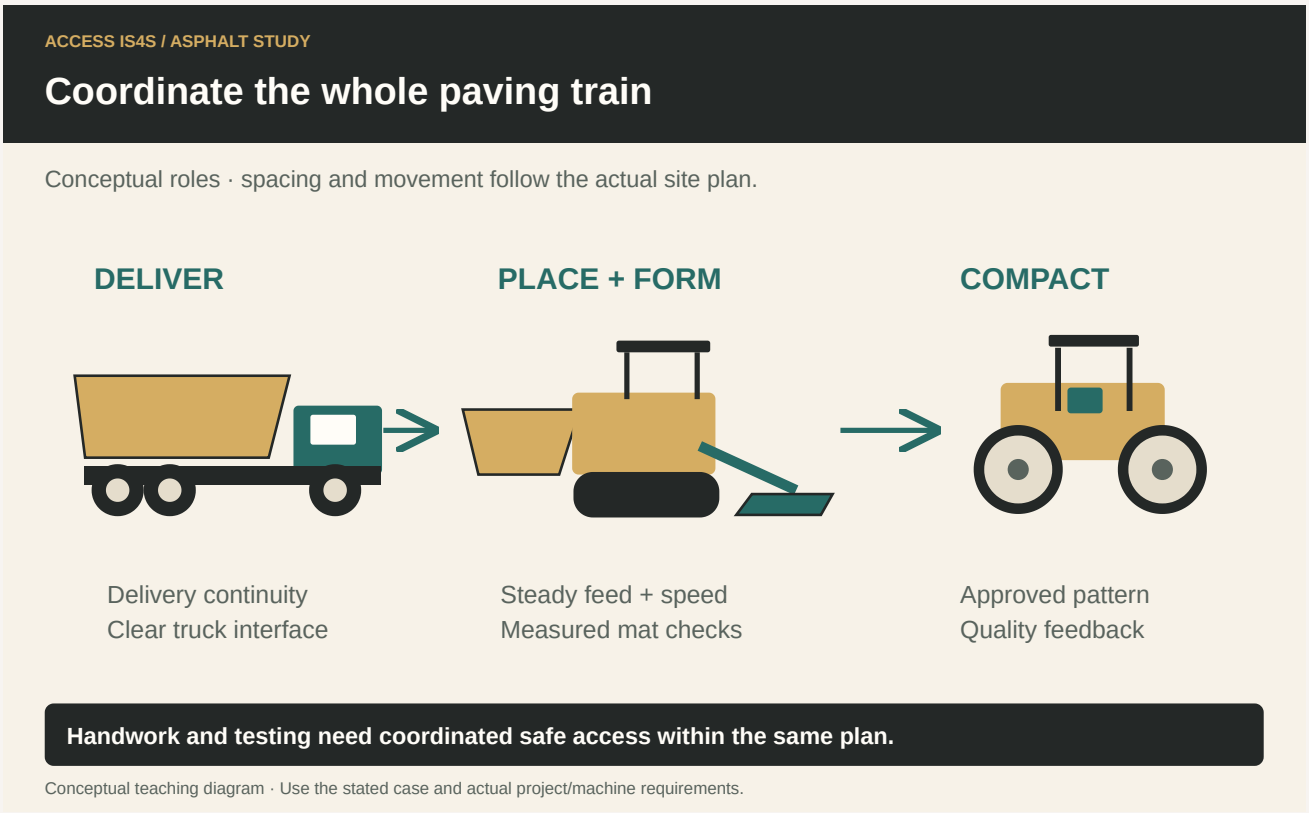
- Standard moves
- Maneuvering the paver
- Position paver
- Simulate paving laying down a mat using 5mm crush and sand mixture or similar compact quality material or asphalt

LEARNING STEP 1

Standard moves

Complete pre-start checks and daily maintenance, identify controls and warnings, and enter using the designated access and three-point contact. Follow the OMM start and travel sequence. Practise forward–stop–reverse–stop on the cleared flat training area, checking surroundings before each movement. Apply the specified parking and transmission controls. Transmission neutral or a control lock is not a substitute for maintenance lockout.

Coordinate the whole paving train



Note. ACCESS IS4S. (2026). Coordinate the whole paving train [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Maneuvering the paver

Manoeuvre smoothly within the machine's operating limits and the instructor's plan. Maintain awareness of space, position, clearance and ground support. In-school learners stay on the approved flat area. Hills, slopes and varied conditions require a separate workplace plan, machine limits and supervised assessment; do not infer a safe slope from a generic diagram. Stop when visibility or support is uncertain.

LEARNING STEP 3

Position paver

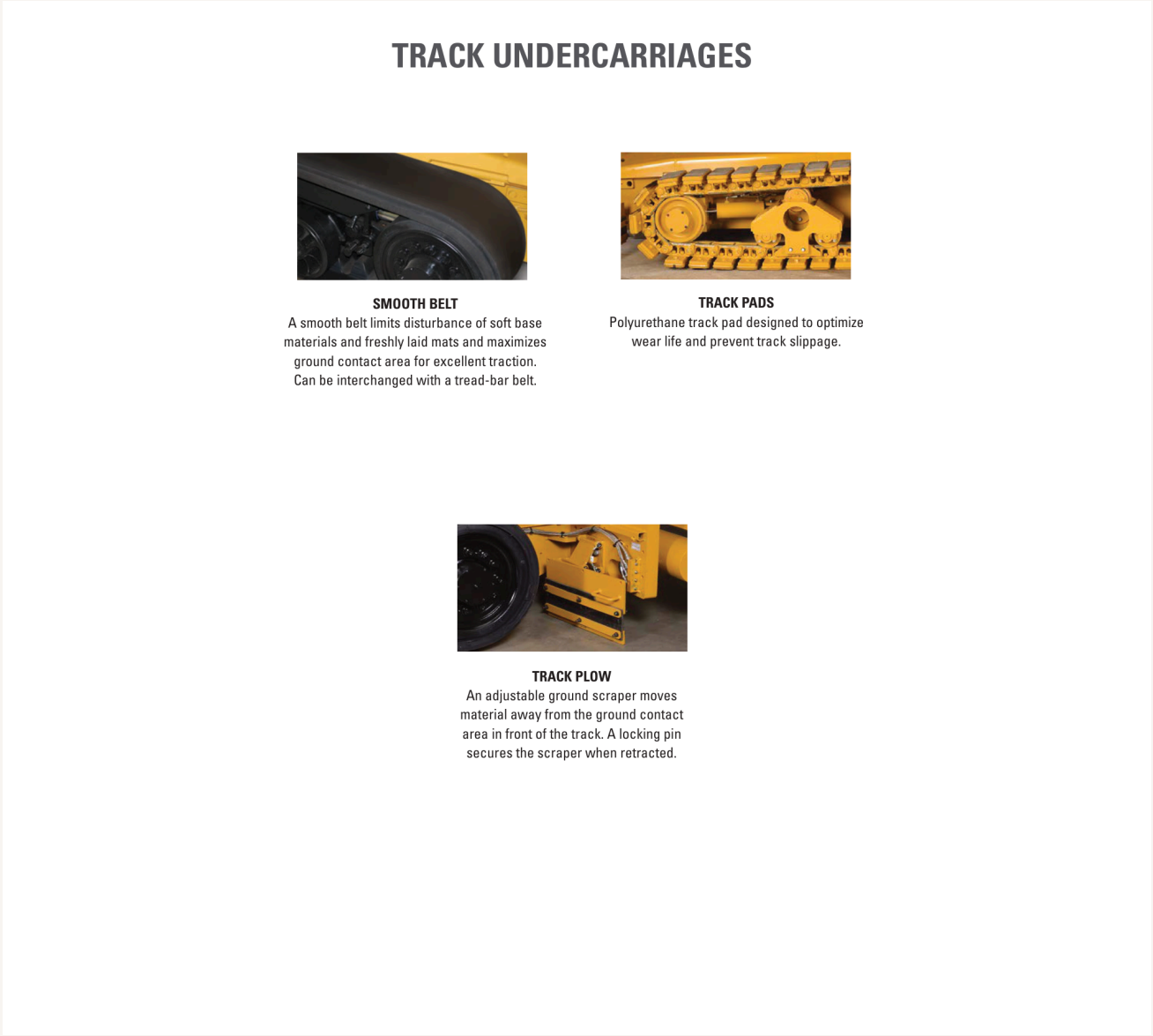
Position the paver for different starting scenarios, confirming line, width, joint, truck approach and screed readiness. Coordinate hopper opening/closing, material-flow settings and truck guidance through the actual controls and agreed signals. Observe the whole movement envelope. The instructor sets time and quantity appropriate to conditions; speed is not rewarded at the expense of control.

LEARNING STEP 4

Simulate paving laying down a mat using 5mm crush and sand mixture or similar compact quality material or asphalt

For a simulated mat, use only an instructor- and manufacturer-approved training material and method; a sand/crushed-material exercise does not reproduce every hot-mix behaviour. For asphalt, follow the approved mixture, temperature, placement and rolling plan. Maintain the line and steady process, communicate with screed and ground crew, monitor performance, and clean/shut down using the safe procedure. Record classroom and workplace observations separately.

Paver contact with the ground



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 19). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

Worked example · Flat practice is a specific achievement

A learner completes a controlled flat-pad pass with simulated material. The next job involves a steep access road and hot mix.

Think it through.

Record the demonstrated flat-pad skills. Plan the additional machine/site-specific preparation and supervised workplace assessment for grades, slopes and hot-mix conditions. Do not label the first exercise as proof of unrestricted operation.

Try it and explain your thinking

List five observations an instructor should record during a flat-pad pass.

List five observations an instructor should record during a flat-pad pass.

Skills you use: Writing, Reading.

A fictional 36 m practice pass takes 12 min

A fictional 36 m practice pass takes 12 min. Calculate average travel speed and one reason not to compare learners by it alone.

Skills you use: Numeracy, Problem solving.

During the pass, a partner reports a person entering the work area

During the pass, a partner reports a person entering the work area. Demonstrate the response.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Precheck, maintain, start and monitor the actual machine.	0 / 1 / 2	Evidence/help: _____
Enter/exit safely and perform flat forward/stop/reverse/stop with correct parking controls.	0 / 1 / 2	Evidence/help: _____
Operate hopper, transmission and flow controls under supervision.	0 / 1 / 2	Evidence/help: _____
Position and guide the truck, follow the line and coordinate the screed/ground crew.	0 / 1 / 2	Evidence/help: _____
Place an approved simulated or asphalt mat, then clean and shut down.	0 / 1 / 2	Evidence/help: _____
Record additional workplace grades/slopes/conditions separately.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I7, printed p. 78; source PDF pp. 80–81.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I8

Follow shut down procedures

A controlled shutdown preserves the paver’s condition and leaves a clear handover. The process includes parking, heat and energy controls, inspection and security—not simply switching off the engine.

Shut-down procedures

LEARNING STEP 1

Shut-down procedures

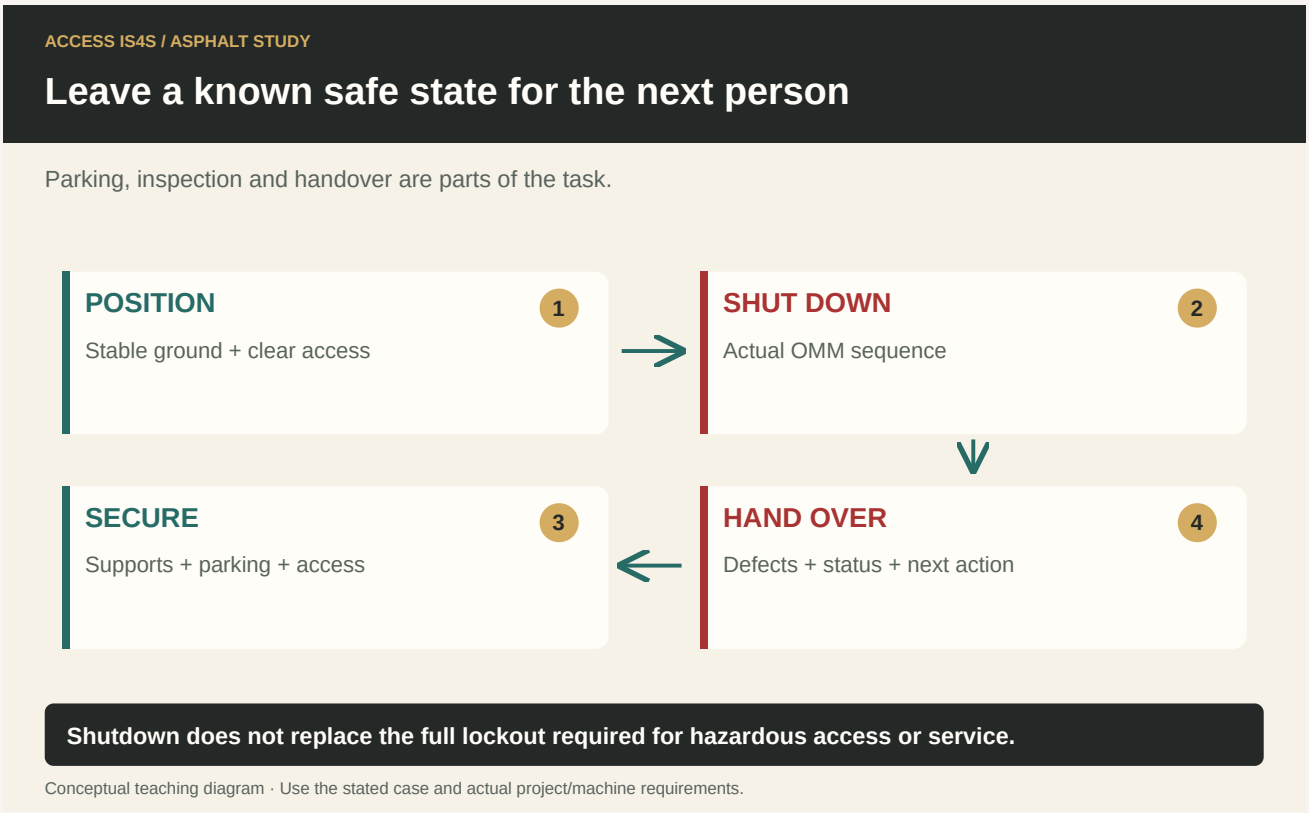
Select the designated stable parking location with service access and clear public/emergency routes. Consider the possibility that the machine will not restart. Use level ground where required for the procedure and fluid checks. Coordinate completion of material handling and screed work before the final stop.

Follow the matching OMM sequence for stopping working functions, positioning the screed and attachments, parking controls, required cooling and engine shutdown. Do not transfer a cooldown time or control sequence from a different model. Use specified supports, locks and chocks. A normal shutdown does not by itself authorize maintenance access: apply the full lockout procedure when the next task requires it.

Inspect from safe access areas for leaks, damage, buildup and missing guards. Keep away from hot or pressurized systems and moving/unsupported parts. Record machine ID, meter, parking location, defects and required service. Secure keys, panels and detachable items, then communicate the final status to the responsible person and next shift.

The practical checklist uses the actual machine procedure. Safety-critical parking, support and energy-control items must all pass. The outline’s 80% checklist threshold cannot compensate for an unsafe final state. Repeat the required practice after feedback and record the new attempt separately.

Leave a known safe state for the next person



Note. ACCESS IS4S. (2026). Leave a known safe state for the next person [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LeeBoy 8680 screed locks and hopper lock valves

Screed Locks

This machine is provided with screed lock mechanisms as added safety and stability during transport. A screed lock knob is located on each side of the screed on the paver frame. **(Figure 4-1)**

- Pull the knob outward to release the screed lock.
- Push the knob in and down to lock the screed in place.

⚠CAUTION Ensure locks are in the unlocked position before raising or lowering the screed.

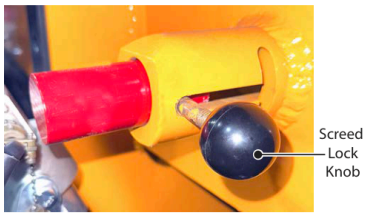


Figure 4-1. Screed Lock (Left Side Shown)

Hopper Lock Valves

The hopper cylinder valves can be used to lock the hopper wings when they are open or closed. **(Figure 4-2)** Be sure to unlock the hopper valves before operation!

- Turn the valve counterclockwise to unlock the hopper wings.
- Turn the valve clockwise to lock the hopper wings.

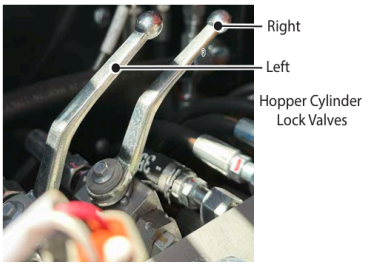


Figure 4-2. Hopper Cylinder Lock Valve

⚠CAUTION Ensure the hopper cylinder is unlocked before extending or retracting the hopper wings.

Note. LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP, p. 4-4; PDF p. 72). Serial 365152 and above. Faithful crop; use the complete matching procedure. Full source context.

Worked example · An incomplete handover

The paver is parked correctly, but the operator noticed a leak and left without recording it.

Think it through.

The shutdown/handover is incomplete. Record the leak observation, location and machine status, notify the responsible person and arrange the required hold/inspection. The next operator needs the information before starting.

Try it and explain your thinking

Explain shutdown versus lockout.

Explain shutdown versus lockout.

Skills you use: Reading, Communication.

A ten-item assessment gives 17/20 but a required screed-support check failed

A ten-item assessment gives 17/20 but a required screed-support check failed. State the result.

Skills you use: Numeracy, Problem solving.

Prepare a concise next-shift note.

Prepare a concise next-shift note.

Skills you use: Writing, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Choose and secure the approved parking location.	0 / 1 / 2	Evidence/help: _____
Complete model-specific shutdown and component support.	0 / 1 / 2	Evidence/help: _____
Inspect safely and hand over defects and status.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I8, printed p. 80; source PDF pp. 82–82.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline I9

Transport paver

Moving a paver between sites requires a verified transport configuration and a coordinated loading plan. The fitted screed and attachments can change width, mass and securement requirements.

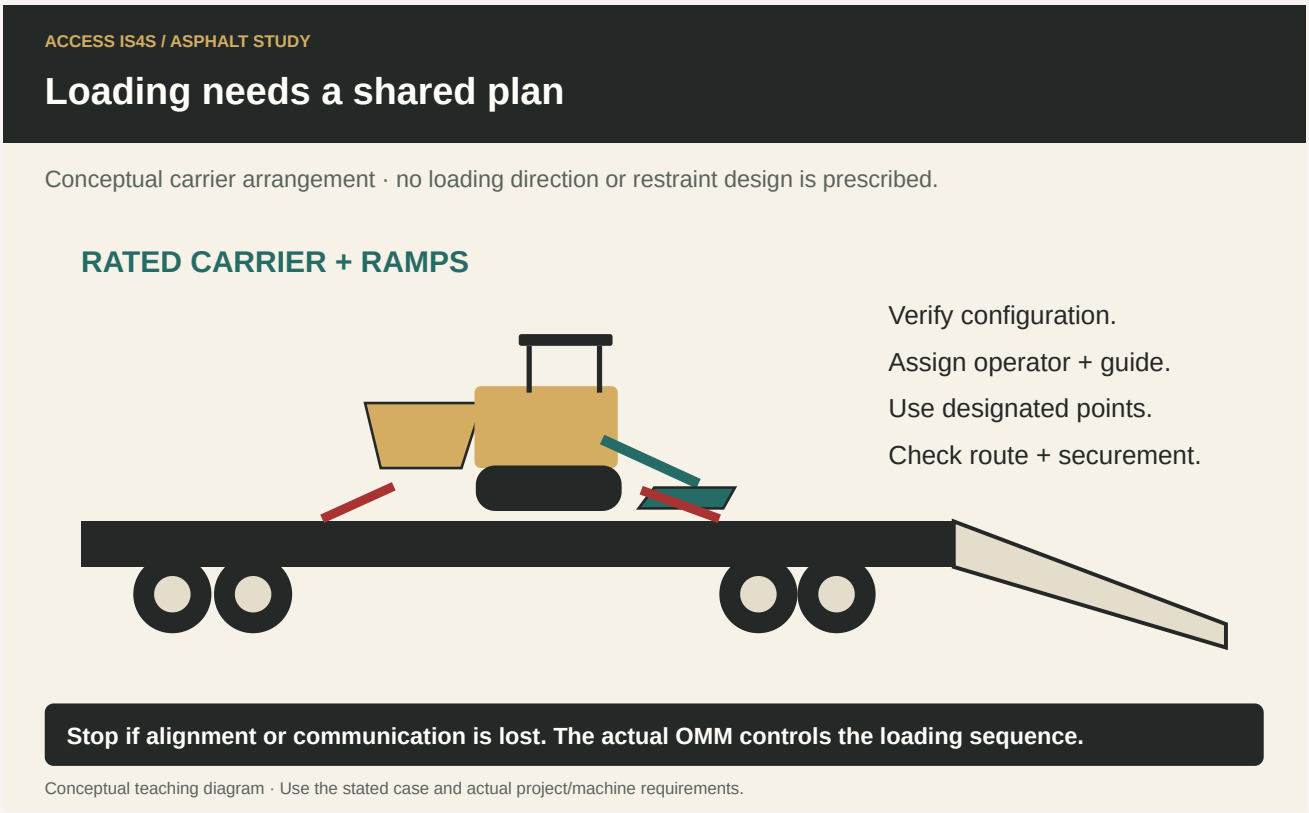
- Compliance with wide load regulations
- Roles and responsibilities
- Prepare machine for transport
- Loading and unloading procedures

LEARNING STEP 1

Compliance with wide load regulations

Measure actual transport dimensions and confirm the load, route and applicable BC commercial transport requirements. Signs, lighting, permits, pilot cars and travel restrictions depend on the actual movement; they are not triggered by a single classroom rule. Removing extensions may change width but requires the approved handling and securement method. The carrier confirms the current requirements before departure.

Loading needs a shared plan



Note. ACCESS IS4S. (2026). Loading needs a shared plan [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Roles and responsibilities

Assign a competent machine operator, carrier/driver and ground guide as required. The driver/carrier manages transport compliance and securement within their responsibility; the designated operator follows the machine loading procedure. Do not assume the paver operator is always included or excluded. Everyone must know the stop signal, communication method and final release responsibility.

LEARNING STEP 3

Prepare machine for transport

Follow the OMM transport configuration for screed, extensions, augers, cut-offs, guards and loose equipment. Use approved lifting/rigging for removed components and designated tie-down points. The LeeBoy 8680 example gives specific preparation and loading instructions in printed pages 4-64–4-65; those instructions apply to the stated model/serial scope, not every paver. Read the complete procedure before a practical demonstration.

LEARNING STEP 4

Loading and unloading procedures

Check carrier/ramp capacity, alignment, support, traction and the clear movement envelope. Use the specified loading direction and controlled speed, with a guide positioned safely. Stop if alignment or communication is lost. After loading, position/support the machine, apply required parking controls and secure using rated components and the applicable National Safety Code Standard 10/BC requirements. Control the machine before releasing restraints for unloading. A time target never justifies rushing a critical step.

LeeBoy 8680 loading arrangement

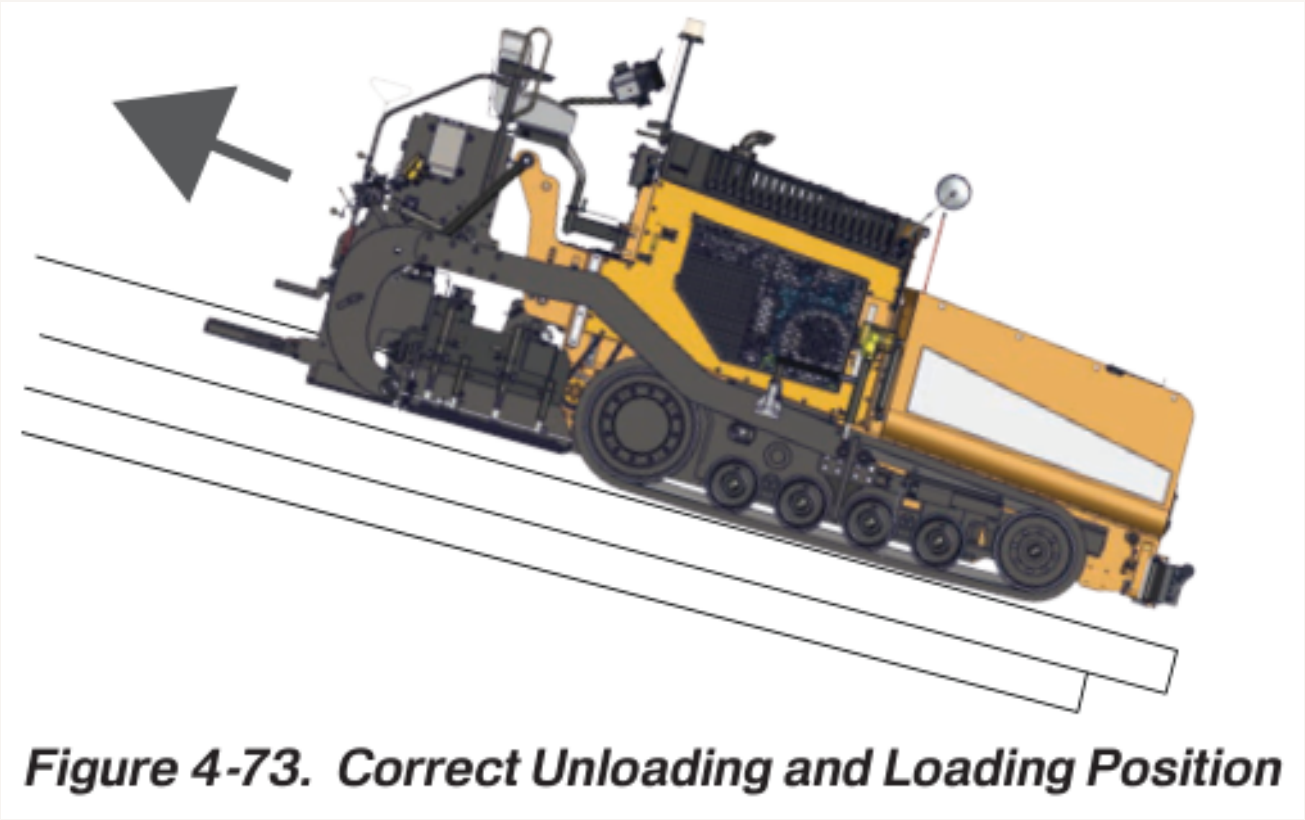


Figure 4-73. Correct Unloading and Loading Position

Note. LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP, p. 4-64; PDF p. 132). Serial 365152 and above. Faithful crop; use the complete matching procedure. Full source context.

Worked example · Width changes with configuration

A fictional machine measures 2.50 m with transport equipment retracted and 3.10 m with a fitted extension assembly.

Think it through.

The actual configured width differs by 0.60 m. Verify the required transport configuration and current route/permit conditions with the carrier. The smaller dimension cannot be used on paperwork while the larger configuration remains fitted.

Try it and explain your thinking

Name four checks before the machine touches the ramp.

Name four checks before the machine touches the ramp.

Skills you use: Reading, Problem solving.

A base machine is 8,600 kg and fitted items add 740 kg

A base machine is 8,600 kg and fitted items add 740 kg. Find the load before separate cargo.

Skills you use: Numeracy.

Write a handover when loading is delayed by an unidentified tie-down point.

Write a handover when loading is delayed by an unidentified tie-down point.

Skills you use: Writing, Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Verify dimensions, transport rules and assigned roles.	0 / 1 / 2	Evidence/help: _____
Prepare screed/attachments to the actual OMM.	0 / 1 / 2	Evidence/help: _____
Demonstrate controlled loading preparation and rated securement checks.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, I9, printed p. 81; source PDF pp. 83–83.
- Canadian Council of Motor Transport Administrators. (June 2013). National Safety Code Standard 10: Cargo Securement. Open publication.
- Canadian Council of Motor Transport Administrators. (May 2016). Cargo Securement Standard: Interpretations and Guidance. Open publication.
- Province of British Columbia. (Web snapshot: October 30, 2026). Cargo Securement. Open publication.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Keep delivery and mat records consistent

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional approved area is 50 m × 3.2 m with a 45 mm compacted target. A verified trial relationship is loose/compacted = 1.24 for this exercise. The delivery register contains ticket 81 for 12 t and ticket 82 for 11 t, but a draft totals 35 t by counting ticket 81 twice. No density for estimating required mass is supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

The coordinator asks for an audio handover as well as an editable table and PDF.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.